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CONDENSED MATTER & LOW TEMPERATURE PHYSICS 2026

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Book of Abstracts

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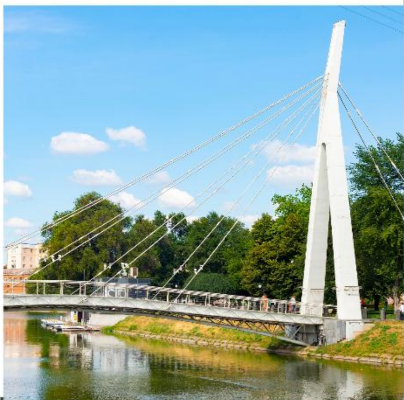

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This book collects 245 peer-reviewed reports presented at the VI International Conference "Condensed Matter and Low Temperature Physics" 2026. These materials present the studies of modern aspects of condensed matter and low temperature physics including electronic properties of conducting and superconducting systems, magnetism and magnetic materials, optics, photonics and optical spectroscopy, quantum liquids and quantum crystals, cryocrystals, nanophysics and nanotechnologies, biophysics and physics of macromolecules, materials science, theory of condensed matter physics, technological peculiarities of the instrumentation for physical experiments, and related fields.

The book will be useful to undergraduate, postgraduate students, and researchers in the field of Physics and Condensed matter physics.

Ця книга зібрала 245 доповідей, представлених на VI Міжнародній конференції "Condensed Matter and Low Temperature Physics" 2026 року. Дані матеріали представляють дослідження у галузі сучасних аспектів фізики конденсованого стану та низьких температур, у тому числі електронні властивості провідних та надпровідних систем, магнетизм, оптику, фотоніку та оптичну спектроскопію, квантові рідини та квантові кристали, кріокристали, нанофізику та нанотехнології, біофізику та фізику макромолекул, матеріалознавство, теорію фізики конденсованого стану, технологічні особливості обладнання для фізичних експериментів та суміжні галузі.

Книга призначена для студентів, аспірантів та дослідників у галузі загальної фізики та фізики конденсованого стану.

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Optimization of thermal transport properties in silicon structures with inhomogeneous porosity: experiment and machine learning

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Understanding heat transport in porous and multilayer silicon structures is of great importance for modern micro- and nanoelectronics, thermoelectric systems, and thermal insulation technologies, where accurate control of thermal conductivity is essential for device reliability. To address these challenges, this study employs a comprehensive approach combining experimental measurements, molecular dynamics (MD) simulations, and machine learning.

Firstly, the thermal and transport properties were investigated using the photoacoustic (PA) method. This non-destructive technique is based on the generation of acoustic waves resulting from periodic heating of a sample by modulated light. The thermophysical parameters, specifically thermal diffusivity and thermal conductivity, were determined from the frequency-dependent amplitude and phase of the PA signal.

In parallel, MD simulations were performed using the Green–Kubo method, which involves integrating the heat current autocorrelation function. Furthermore, a Physics-Informed Neural Network (PINN) was employed to solve the 2D Fourier heat conduction equation for porous models generated using the Quartet Structure Generation Set (QSGS) algorithm. The figure presents the results obtained for a homogeneous structure at room temperature. A strong correlation is observed between the thermal conductivity values obtained using the different approaches. The slightly lower thermal conductivity predicted by MD simulations can be attributed to finite-size effects that limit the contribution of long-wavelength phonons, as well as to the finite simulation time. As porosity increases, phonon scattering at pores becomes the dominant heat-transport mechanism, reducing the discrepancy between the methods.

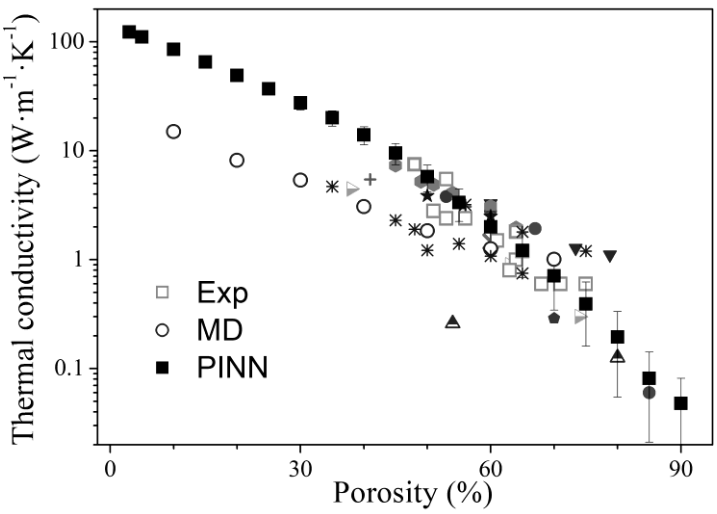

Fig. 1. Thermal conductivity of porous silicon as a function of porosity obtained using various methods. Experimental data reported in the literature are also included for comparison.

Finally, the study evaluated multilayer silicon structures with varying porosity. Both experimental and PINN-based data deviate from the effective medium approximation, with discrepancies increasing alongside the porosity contrast between layers. The interfacial thermal resistance was quantitatively estimated, providing essential data for the design of advanced thermal insulation components.

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